



mains supply is available, white LED is off—irrespective of the position of

dots in the circuit. Now join nearest ends of the coils together, as shown in

switch S1. When AC mains supply is off, switch S1 should be closed to make white LED glow and opened to switch it off.

For the winding on toroidal core, take a small toroidal core (for example, one with 10mm outer dia and 5mm height). Wrap a 36SWG enamelled copper wire around the toroidal core to make 10 turns. Repeat the winding with the second piece of 36SWG enamelled copper wire. The two windings (L1 and L2) should be in opposite directions. The polarities of the windings are shown by the



Fig. 6. This forms the centre pole which is connected to positive terminal of the battery. Before using the circuit, verify the voltages are as per table.

Caution: The circuit operates off live 230V AC mains, so take all the necessary safety precautions while building and using this circuit to avoid any nasty electric shock. ●

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